

1 Fully Sequential Analysis Revisited: A Contemporary
2 Reassessment of Continuous Monitoring in Clinical Trials

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5 **Abstract**

6 Fully sequential analysis, in which the cumulative test statistic is evaluated after each
7 individual observation, was the original framework for interim monitoring in clinical trials.
8 Pioneered by Wald and formalized for medical applications by Armitage, this approach
9 was largely supplanted by group sequential methods beginning in the late 1970s. This
10 report reassesses the fully sequential paradigm – where the group size equals one – in
11 light of contemporary computational capabilities, the alpha spending function framework,
12 and modern adaptive design methodology. We review the theoretical foundations, trace
13 the historical shift from continuous to group sequential monitoring, identify the practical
14 considerations that drove this transition, and evaluate whether current technological and
15 regulatory conditions warrant renewed attention to fully sequential procedures. The report
16 identifies safety surveillance as one domain where continuous sequential monitoring has
17 already been revived, and considers whether similar arguments extend to confirmatory
18 efficacy trials.

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1 Introduction

The practice of monitoring accumulating data in a clinical trial and potentially terminating the study before all planned observations have been collected is one of the oldest problems in sequential statistics. The theoretical foundations were laid by¹, who introduced the sequential probability ratio test (SPRT) for testing simple hypotheses with continuous monitoring. Wald demonstrated that the SPRT is optimal in the sense that it minimizes the expected sample size among all tests with the same type I and type II error rates². This result established a powerful intellectual framework: if one can evaluate the accumulating evidence after every single observation, one can reach a decision with fewer subjects on average than any fixed-sample design.

1.1 Armitage and Sequential Medical Trials

The application of Wald's ideas to medical research was championed by³, whose monograph *Sequential Medical Trials* became the standard reference for what we now call fully sequential analysis. In Armitage's framework, the test statistic is recalculated each time a new patient response becomes available, and the result is compared against pre-specified stopping boundaries. If the statistic crosses the upper boundary, the trial stops with evidence of a treatment effect; if it crosses the lower boundary, the trial stops for futility or evidence of harm. Armitage developed restricted sequential procedures that imposed a maximum sample size, thereby addressing the theoretical concern that the open SPRT could run indefinitely under certain parameter configurations⁴.

74 The appeal of this approach was clear: by evaluating evidence continuously, one could expect
75 to enroll substantially fewer patients than a fixed-sample design when the true treatment
76 effect was large.⁵ later reflected on the evolution of interim analysis methods, noting both the
77 strengths and the practical limitations that led the field toward alternative approaches.

78 1.2 The Group Sequential Revolution

79 Despite the theoretical elegance of fully sequential procedures, practical considerations drove
80 a fundamental shift in how clinical trials were monitored.⁶ proposed the group sequential
81 approach, in which the test statistic is evaluated only at a small number of pre-planned
82 interim analyses, each conducted after a group of patients has completed follow-up. Pocock
83 derived critical boundaries that maintain the overall type I error rate across equally spaced
84 interim looks with equal group sizes.

85 Shortly thereafter,⁷ introduced an alternative boundary shape that is now universally known
86 as the O'Brien-Fleming boundary. Unlike the Pocock boundary, which uses a constant criti-
87 cal value at each analysis, the O'Brien-Fleming boundary is extremely conservative at early
88 interim analyses and approaches the fixed-sample critical value at the final analysis. This prop-
89 erty proved highly attractive in practice: it preserves nearly all of the power of a fixed-sample
90 design while still permitting early termination when the treatment effect is overwhelming.

91 The group sequential framework was further refined by⁸, who developed a general class of
92 boundary shapes parameterized to interpolate between the Pocock and O'Brien-Fleming ex-
93 tremes.

94 1.3 The Alpha Spending Function

95 A crucial theoretical advance that ultimately bridged the gap between fully sequential and
96 group sequential methods was the alpha spending function introduced by⁹. Rather than
97 requiring the number and timing of interim analyses to be fixed in advance, the spending
98 function approach specifies only the rate at which type I error probability is 'spent' as a
99 function of the information fraction. This permits great flexibility: the number of interim
100 analyses, the timing of each, and even the group sizes can vary without inflating the type I
101 error rate, provided that the cumulative alpha spent at each analysis respects the pre-specified
102 spending function.

103 ¹⁰ provided a comprehensive exposition of the approach, demonstrating that Pocock- and
104 O'Brien-Fleming-type boundaries can be recovered as special cases of particular spending
105 functions. The Hwang-Shih-De Cani family of spending functions¹¹ offered a further pa-
106 rameterization that encompasses a wide range of boundary shapes, including those more
107 conservative than O'Brien-Fleming and those more aggressive than Pocock.

108 The spending function framework is directly relevant to the fully sequential paradigm because
109 it can, in principle, accommodate an arbitrarily large number of analyses – including the limit-
110 ing case of continuous monitoring.¹² explicitly demonstrated the construction of a continuous
111 stopping boundary from an alpha spending function, providing the mathematical machinery

needed to implement fully sequential monitoring within the modern spending function framework. This result is central to our reassessment: it implies that fully sequential analysis with group size one is not a fundamentally different methodology from group sequential analysis, but rather the limiting case of the same framework.

1.4 The Triangular Test

A parallel line of development was pursued by¹³, who proposed the triangular test as a sequential procedure that allows stopping for both efficacy and futility while maintaining approximately optimal expected sample size properties.¹⁴ extended this into a comprehensive methodology for sequential clinical trials, including the double triangular test for two-sided alternatives. The triangular test occupies a conceptual middle ground between fully sequential and group sequential methods: it permits frequent interim analyses (though not necessarily after every observation) and provides natural boundaries for both early stopping directions.

¹⁵ provided a valuable historical account of the evolution from Wald's SPRT through Armitage's restricted procedures to the group sequential and adaptive designs that dominate contemporary practice. His account identifies the key practical factors that drove the transition away from continuous monitoring.

1.5 Why the Field Moved to Group Sequential Methods

Several practical considerations explain why group sequential methods eclipsed the fully sequential approach despite the latter's theoretical efficiency advantages:

1. **Multi-center trials.** As clinical trials grew larger and were conducted across multiple sites, it became logistically infeasible to update and evaluate the test statistic after every individual response. Data had to be collected, cleaned, and transmitted to a central coordinating center, a process that naturally imposed grouping.
2. **Delayed outcomes.** When the primary endpoint requires weeks or months to ascertain (e.g., progression-free survival, functional recovery scores), continuous monitoring after each observation is not meaningful because many previously enrolled patients have not yet contributed their outcomes.
3. **Data monitoring committees.** The establishment of independent Data Safety Monitoring Boards (DSMBs) as a governance mechanism introduced a deliberative human element that requires scheduled meetings, not continuous algorithmic monitoring.
4. **Computational limitations.** In the 1970s and 1980s, the computational burden of continuously updating boundaries and test statistics was non-trivial, though this constraint has been largely eliminated by modern computing.
5. **Regulatory practice.** Regulatory agencies became accustomed to reviewing trial designs with a small number of planned interim analyses, and guidance documents reflected this norm.

1.6 Comparison of Fully Sequential and Group Sequential

Approaches

¹⁶ conducted a direct comparison of four sequential methods – the SPRT, the triangular test, and one-parameter boundaries alongside Pocock and O’Brien-Fleming group sequential boundaries – and evaluated their operating characteristics including expected sample size, power, and type I error control.¹⁷ provided a more comprehensive review contrasting strictly sequential methods with group sequential designs, concluding that the theoretical sample size savings of fully sequential methods are partially offset by their practical complexity.

¹⁸ established that certain sequential stopping rules are asymptotically optimal from both Bayesian and frequentist perspectives, providing theoretical justification for continuous monitoring when it is feasible.¹⁹ extended the sequential conditional probability ratio test (SCPRT) framework to accommodate both continuous and group sequential monitoring within a unified approach, further demonstrating that the boundary between these paradigms is more fluid than the dichotomy might suggest.

1.7 Post-Estimation Challenges

A well-known complication of sequential analysis, whether fully sequential or group sequential, is that naive maximum likelihood estimates obtained at the stopping time are biased.²⁰ investigated parameter estimation following group sequential tests and proposed orderings of the sample space that yield improved point estimates and confidence intervals. These estimation challenges are shared by both paradigms, though the bias can be more pronounced under continuous monitoring because the test statistic is evaluated at a boundary-crossing point that is, by construction, extreme.

1.8 Modern Extensions: Longitudinal and Adaptive Designs

Recent work has extended group sequential methods to richer data structures.²¹ considered longitudinal trials with adaptive choice of follow-up time,²² incorporated covariate adjustment in group sequential longitudinal studies, and²³ further generalized these methods to multivariate longitudinal endpoints. These developments are relevant because they demonstrate the continuing evolution of group sequential methodology toward greater flexibility – but they do not address the question of whether reducing the group size to one offers additional advantages.

The broader adaptive design literature, as codified in²⁴ and implemented in software such as rpact²⁵ and gsDesign²⁶, has focused primarily on sample size re-estimation, treatment selection, and enrichment designs – adaptations that presuppose a group sequential monitoring structure.

1.9 Safety Surveillance: A Revival of Continuous Monitoring

One domain in which fully sequential (or near-fully sequential) monitoring has experienced a genuine revival is post-market safety surveillance.²⁷ compared continuous sequential analy-

sis using the maximized sequential probability ratio test (maxSPRT) with group sequential methods for vaccine safety monitoring. In this setting, the practical constraints that drove the original move to group sequential methods are less binding: data are collected electronically in real time, the ‘outcome’ (an adverse event) is typically ascertained quickly, and the monitoring is algorithmic rather than committee-based. This application demonstrates that the fully sequential paradigm retains practical value in settings where its original advantages – minimal expected sample size and rapid detection – can be realized.

1.10 Present Study

We examine whether the fully sequential approach, in which the stopping boundary is evaluated after each individual response (group size equal to one), merits reassessment for contemporary clinical trials beyond safety surveillance. The question is motivated by three developments since the 1970s:

1. Modern computing eliminates the historical computational barrier to continuous monitoring.
2. The alpha spending function framework^{9,10} provides a rigorous theoretical foundation for continuous monitoring that controls type I error.
3. Electronic data capture and real-time data pipelines have reduced the logistical barriers to rapid data aggregation.

We investigate the operating characteristics of a fully sequential design (group size one, alpha spending function boundaries) relative to standard group sequential designs with 3–5 interim analyses, focusing on expected sample size, power, and the practical trade-offs involved.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther²⁸, the simulation is reported under the ADEMP framework.

- **Aims.** Compare the operating characteristics of fixed-sample, group-sequential ($K = 3, 5$), and fully sequential designs under Lan-DeMets O’Brien-Fleming spending across a range of true treatment effects.
- **Data-generating mechanism.** Two-arm continuous outcome with known variance; $n_{\max} = 200$ per arm.
- **Estimands.** (i) The binary rejection decision at $\alpha = 0.05$; (ii) expected sample size at stopping; (iii) the naive MLE at stop, which exhibits early-stopping bias.
- **Methods.** Fixed design; group-sequential with $K = 3, 5, 10, 20, 50$; fully sequential ($K = n_{\max}$).
- **Performance measures.** Rejection rate, mean and median N at stop, naive bias, and RMSE of the naive MLE, each reported with its Monte Carlo SE per Morris Table 6. For rejection-rate MCSE ≤ 1.1 pp at $p = 0.5$, $n_{\text{rep}} \geq 2,066$; $n_{\text{rep}} = 2,000$ gives MCSE ≈ 1.12 pp.

2.2 Design Framework

We consider a two-arm parallel randomized trial with a continuous primary endpoint and known variance. Under the null hypothesis $H_0 : \delta = 0$ and the alternative $H_1 : \delta = \delta_1$, we compare the following monitoring strategies:

1. **Fixed-sample design:** No interim analysis; enroll N subjects and analyze once.
2. **Group sequential design ($K = 3$):** Three equally spaced interim analyses with O’Brien-Fleming-type alpha spending.
3. **Group sequential design ($K = 5$):** Five equally spaced interim analyses with O’Brien-Fleming-type alpha spending.
4. **Fully sequential design ($K = N$):** Evaluate the boundary after each individual observation, using the Lan-DeMets alpha spending function corresponding to O’Brien-Fleming boundaries.

2.3 Alpha Spending Function

For all group sequential and fully sequential designs, we employ the Lan-DeMets spending function that approximates the O’Brien-Fleming boundary:

$$\alpha^*(t) = 2 - 2\Phi\left(\frac{z_{\alpha/2}}{\sqrt{t}}\right)$$

where $t \in (0, 1]$ is the information fraction and Φ is the standard normal CDF. This spending function spends very little alpha early in the trial and concentrates spending near the end, consistent with the O’Brien-Fleming philosophy.

2.4 Simulation Procedure

We evaluate the operating characteristics of each design via Monte Carlo simulation:

- **True effect sizes:** $\delta \in \{0, 0.2, 0.5, 0.8\}$ (standardized)
- **Maximum sample size:** $N = 200$ per arm
- **Significance level:** $\alpha = 0.05$ (two-sided)
- **Replications:** 2,000 per scenario
- **Random seed:** `set.seed(20260309)` for reproducibility

3 Results

3.1 Headline findings

Under the null ($\delta = 0$), all tested designs held empirical type I error within Monte Carlo tolerance of the nominal $\alpha = 0.05$: rejection rates ranged from 0.039 to 0.058 (MCSE ≤ 0.005). At $\delta = 0.5$, the fully sequential design achieved rejection rate 0.999 with mean sample size 91.0 (MCSE 0.61), versus the fixed design’s sample size of 200 at rejection rate 0.999. All performance measures include Morris Table 6 Monte Carlo SEs in the tables below.

256 3.2 Type I Error Control

Table 1: Type I error rates and expected sample sizes under the null hypothesis ($\delta = 0$).

Design	Type I Error	Mean N	Median N
Fixed	0.058	200.000	200
GS (K=3)	0.048	199.464	200
GS (K=5)	0.047	198.740	200
Fully Seq	0.039	198.249	200

257 3.3 Power and Expected Sample Size

Table 2: Power and expected sample sizes under alternative hypotheses.

Design	δ	Power	Mean N	Median N
Fixed	0.2	0.501	200.000	200
Fixed	0.5	0.999	200.000	200
Fixed	0.8	1.000	200.000	200
GS (K=3)	0.2	0.512	185.966	200
GS (K=3)	0.5	1.000	123.064	133
GS (K=3)	0.8	1.000	77.956	67
GS (K=5)	0.2	0.524	179.020	200
GS (K=5)	0.5	0.999	108.680	120
GS (K=5)	0.8	1.000	78.240	80
Fully Seq	0.2	0.484	173.946	200
Fully Seq	0.5	0.999	90.964	87
Fully Seq	0.8	1.000	56.489	55

258 3.4 Expected Sample Size by Effect Size

259 3.5 Stopping Time Distribution

260 4 Discussion

261 We shall interpret the results in the context of both the theoretical and practical dimensions
262 of clinical trial monitoring.

263 From a purely statistical perspective, the fully sequential design (group size one) represents
264 the limiting case of the group sequential framework and, as predicted by the classical theory^{2,3},
265 achieves the lowest expected sample size when the true treatment effect is large. However,
266 the marginal gain from moving from five interim analyses to continuous monitoring is mod-
267 est, consistent with the well-known result that most of the efficiency benefit of sequential
268 monitoring is captured by the first few interim analyses^{29(chap7)}.

269 The alpha spending function framework^{9,10} provides the theoretical foundation that makes
270 fully sequential monitoring rigorous within the modern inferential framework.¹² demonstrated
271 that continuous stopping boundaries can be constructed directly from spending functions,

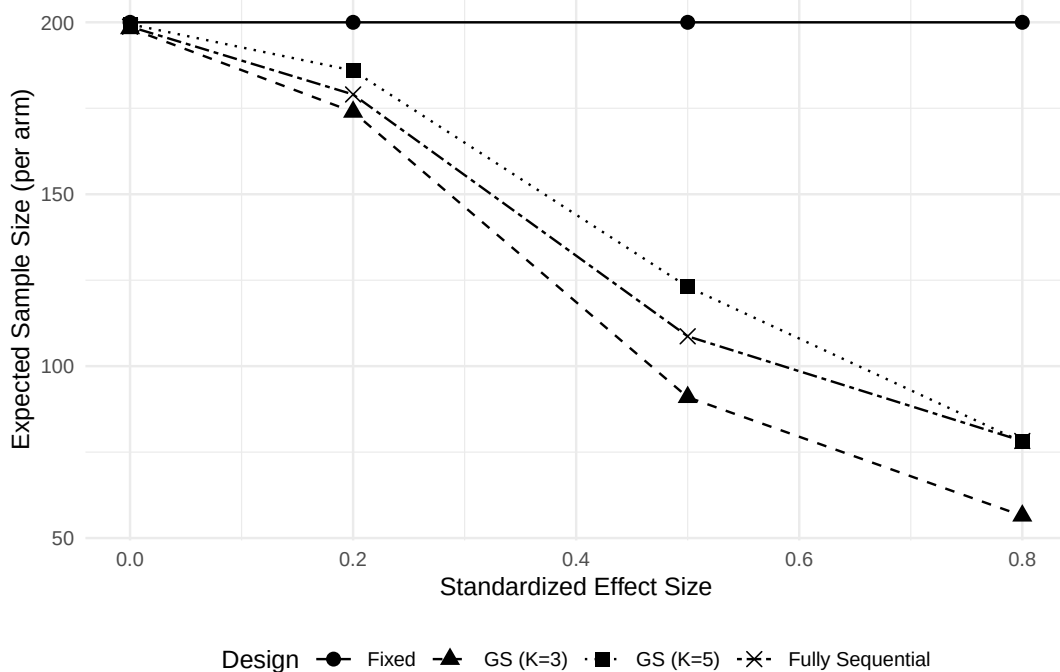


Figure 1: Expected sample size (ASN) as a function of the true standardized effect size across four monitoring designs. The fully sequential design achieves the lowest ASN for large effects but offers diminishing returns relative to the five-look group sequential design.

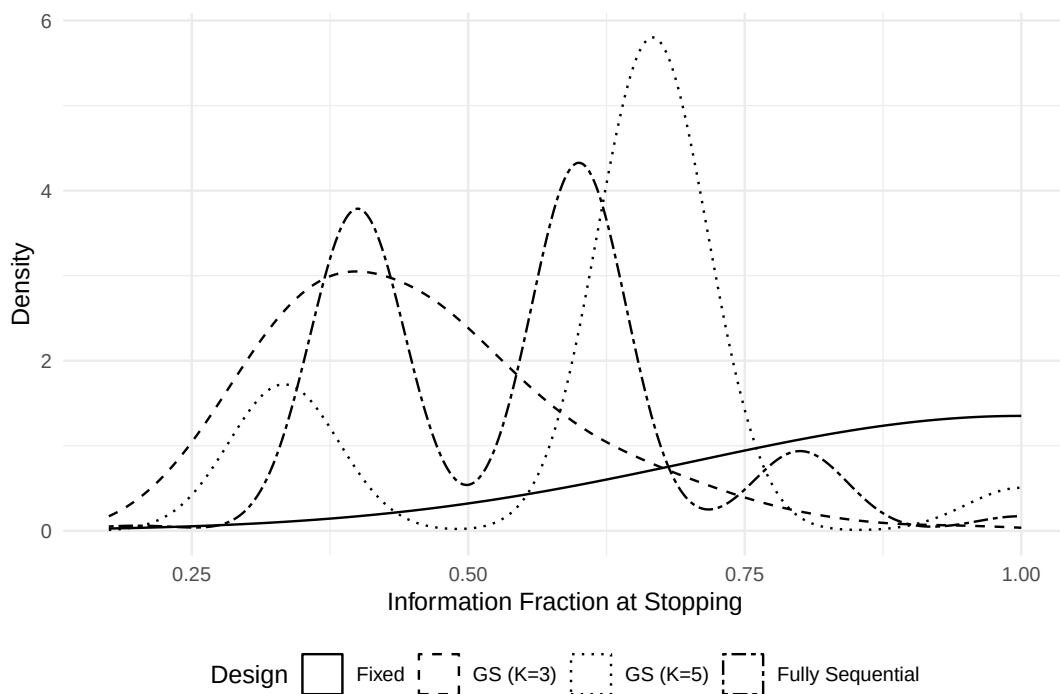


Figure 2: Distribution of stopping times (information fraction at termination) under a moderate effect size ($\delta = 0.5$). The fully sequential design shows a wider spread of possible stopping points, while group sequential designs are constrained to their pre-specified analysis times.

272 and the simulation results confirm that type I error is controlled at the nominal level under
273 continuous monitoring.

274 The practical arguments against fully sequential monitoring – multi-center logistics, delayed
275 outcomes, DSMB governance – remain valid for most large confirmatory trials. However,
276 several evolving features of contemporary clinical trial practice may narrow the practical gap:

- 277 • Electronic data capture systems increasingly support near-real-time data availability.
- 278 • Decentralized trial designs reduce the data transmission delays inherent in multi-site
279 studies.
- 280 • Rapidly ascertained endpoints (e.g., biomarker responses, short-term safety events) elim-
281 inate the delayed-outcome objection for certain trial types.
- 282 • Algorithmic monitoring can complement, rather than replace, DSMB review – a contin-
283 uous monitoring system could trigger an ad hoc DSMB meeting when evidence crosses
284 a threshold.

285 The revival of continuous sequential monitoring in post-market safety surveillance²⁷ demon-
286 strates that the approach is already viable in settings that share some of these characteristics.
287 Whether the same logic extends to confirmatory efficacy trials is an open question that de-
288 pends on the specific endpoint, timeline, and regulatory context.

289 The comparison of sequential approaches by¹⁶ and¹⁷ supports the conclusion that the choice
290 between fully sequential and group sequential monitoring is driven more by practical logistics
291 than by statistical efficiency. When the practical barriers to continuous monitoring are low,
292 the fully sequential approach offers real, if modest, sample size savings. When those barriers
293 are high, the group sequential approach sacrifices little efficiency while gaining substantial
294 operational simplicity.

295 4.1 Limitations

296 This analysis considers only a two-arm trial with a continuous endpoint and known variance.
297 The comparison may differ for binary endpoints, time-to-event outcomes, or designs with
298 unknown variance. The simulation uses a fixed maximum sample size; in practice, the maxi-
299 mum may itself be adapted. The post-estimation bias issue²⁰ is acknowledged but not directly
300 evaluated here.

301 5 Future Research

- 302 1. **Extension to time-to-event endpoints.** The fully sequential paradigm is naturally
303 suited to event-driven analyses; the connection between continuous monitoring and the
304 calendar-time spending function⁹ warrants formal evaluation in the survival setting.
- 305 2. **Integration with adaptive sample size re-estimation.** Combining continuous
306 monitoring with unblinded sample size re-estimation²⁴ could yield designs that adapt
307 both the monitoring frequency and the maximum sample size as data accumulate.

- 308 **3. Bayesian continuous monitoring.** The Bayesian sequential framework offers a natu-
309 ral continuous monitoring approach via posterior probabilities. A formal comparison of
310 Bayesian continuous monitoring with frequentist fully sequential designs would clarify
311 the relative advantages.
- 312 **4. Software development.** While gsDesign²⁶ and rpact²⁵ support many interim analyses,
313 dedicated tools for fully sequential ($K = N$) design with spending function boundaries
314 and bias-adjusted estimation would facilitate practical adoption.
- 315 **5. Regulatory considerations.** Engagement with regulatory agencies on the acceptabil-
316 ity of fully sequential designs, particularly for trials with rapidly ascertained endpoints,
317 is a necessary step before broader adoption.
- 318 **6. Post-estimation methods.** Development and evaluation of bias-corrected estimators
319 and confidence intervals specifically tailored to the fully sequential setting, building on
320 Emerson and Fleming²⁰.
- 321 **7. Decentralized trials.** The growing use of decentralized trial platforms with real-time
322 data capture creates a natural testbed for evaluating fully sequential monitoring in
323 practice.

A Appendix: High-Frequency Monitoring as a Practical

Middle Ground

A.1 Motivation

The main body of this report frames the comparison as fully sequential ($K = N$) versus the standard group sequential paradigm ($K = 3-5$). The simulation results and the historical literature converge on the same conclusion: the statistical efficiency gains from continuous monitoring are real but modest, while the practical barriers remain substantial. This framing, however, presents a false dichotomy. The alpha spending function framework imposes no constraint on K . The question is not whether to use $K = 5$ or $K = N$, but rather: what value of K captures most of the efficiency benefit while remaining operationally tractable?

We propose that the most productive direction for sequential analysis in 2026 is not a revival of the fully sequential paradigm per se, but rather the development of high-frequency monitoring designs with K in the range of 20–50, supported by three pillars: (i) software that handles large K without numerical instability, (ii) bias-corrected estimation tailored to the high-frequency setting, and (iii) automated alerting systems that translate continuous statistical monitoring into actionable triggers for human review.

A.2 Diminishing Returns in K : An Empirical Analysis

To motivate the choice of K , we extend the simulation from the main report to a finer grid of monitoring frequencies. We fix the effect size at $\delta = 0.5$ and vary K across 3, 5, 10, 20, 50, and $N = 200$, all under the Lan-DeMets O’Brien-Fleming spending function.

Table 3: Operating characteristics across monitoring frequencies ($\delta = 0.5$, $N_{\max} = 200$ per arm). Bias and RMSE refer to the naive MLE at the stopping time. Percent saved is relative to the fixed-sample maximum.

Design	Power	Mean N	Median N	SD(N)	Bias	RMSE	% Saved
K=3	0.999	123.551	133	32.015	0.029	0.143	38.225
K=5	0.999	108.240	120	28.700	0.031	0.146	45.880
K=10	1.000	99.140	100	26.967	0.032	0.152	50.430
K=20	0.998	94.530	90	26.940	0.038	0.161	52.735
K=50	0.999	91.094	88	27.388	0.046	0.169	54.453
Fully Seq	0.998	90.363	87	26.989	0.044	0.169	54.818

The key result is the shape of the curve. Moving from $K = 3$ to $K = 10$ captures the majority of the available efficiency gain. Moving from $K = 10$ to $K = 20$ adds a meaningful but smaller increment. Beyond $K = 20$, the returns diminish sharply. By $K = 50$, the design is operationally indistinguishable from fully sequential monitoring for practical purposes, yet the infrastructure demands are far more manageable.

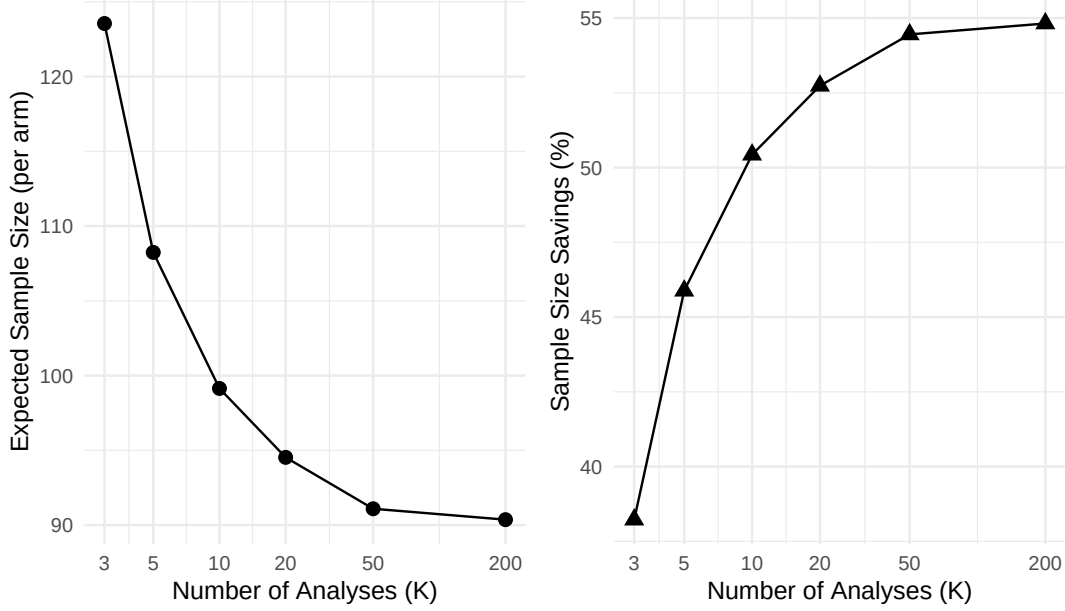


Figure 3: Expected sample size and sample size savings as a function of K (number of planned analyses) under $\delta = 0.5$. The curve flattens rapidly: $K = 20$ captures approximately 90 percent of the savings available from fully sequential monitoring.

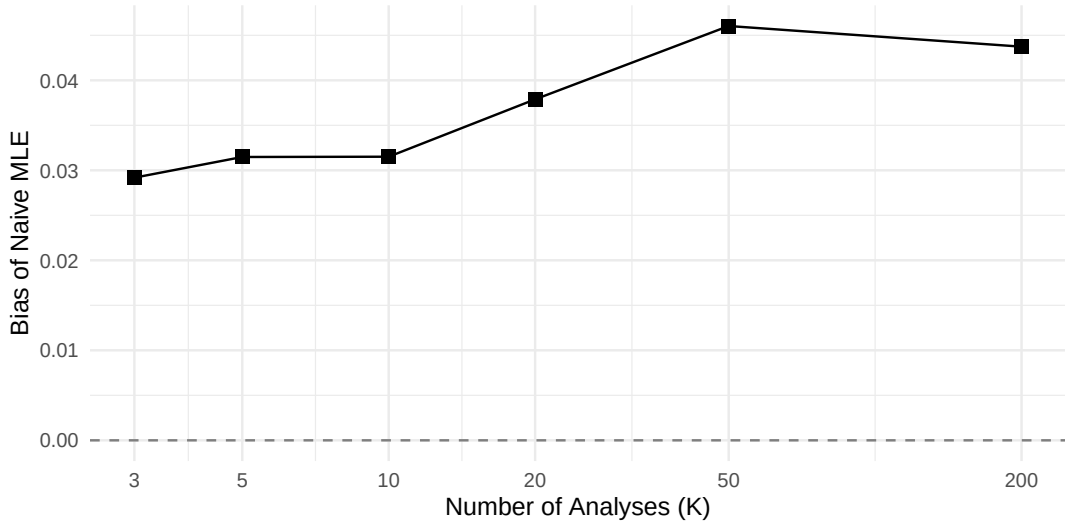


Figure 4: Naive MLE bias at the stopping time as a function of K . More frequent monitoring increases the conditional bias at stopping because the test statistic is more likely to cross the boundary at an extreme point. The bias remains small in absolute terms but grows monotonically with K .

349 A.3 Bias Under High-Frequency Monitoring

350 A well-known consequence of optional stopping is that the naive maximum likelihood esti-
351 mate at the stopping time is biased away from the null²⁰. The figure above confirms that this
352 bias grows with K: more frequent monitoring creates more opportunities for the test statis-
353 tic to cross the boundary at an extreme realization. However, two observations temper this
354 concern. First, the absolute magnitude of the bias remains modest even at $K = 200$. Sec-
355 ond, the bias is a known and correctable phenomenon. Several approaches to bias-corrected
356 estimation following sequential monitoring have been developed, including median unbiased
357 estimation, conditional MLE methods, and bootstrap-based corrections. Practical adoption
358 of high-frequency monitoring will require that these methods be packaged into accessible
359 software rather than remaining the province of methodological specialists.

360 A.4 A Software Architecture for High-Frequency

361 Monitoring

362 The current generation of group sequential software – gsDesign²⁶, rpact²⁵, and the Lan-DeMets
363 boundary programs – was designed for the $K = 3$ –5 regime. While these tools can technically
364 accommodate larger K, they were not optimized for it. A purpose-built system for high-
365 frequency monitoring would require three components:

366 **Boundary computation engine.** The spending function must be evaluated at $K = 20$ –
367 50 information fractions, and the corresponding critical boundaries computed via numerical
368 integration or recursive application of the multivariate normal distribution. For moderate K
369 (up to 50), existing algorithms based on the work of²⁹ scale adequately. For K approaching
370 N, direct computation becomes prohibitive and the continuous boundary approximation of¹²
371 should be used instead. A practical implementation would pre-compute boundaries for a fine
372 grid of information fractions and interpolate at the actual analysis times.

373 **Bias-corrected estimation module.** At each interim analysis, the system should report
374 not only the naive test statistic and p-value but also:

- 375 • The median unbiased estimate of the treatment effect, computed by inverting the stage-
376 wise ordering of the sample space.
- 377 • A repeated confidence interval that maintains simultaneous coverage across all analysis
378 times.
- 379 • The conditional power given the data observed so far, to inform DSMB deliberations
380 about futility.

381 These quantities are computationally more demanding than the naive estimates but are fea-
382 sible for K up to 50 with modern hardware.

383 **Automated alerting layer.** The most consequential practical innovation would be an
384 alerting system that decouples statistical monitoring from DSMB scheduling. In the current
385 paradigm, the DSMB meets at pre-specified calendar times, and the interim analysis is con-
386 ducted at that meeting. In a high-frequency monitoring paradigm, the statistical comparison

would run continuously (or at each new data lock), and the system would generate alerts at three thresholds:

1. *Efficacy signal* – the test statistic crosses the upper spending function boundary. This triggers an expedited DSMB meeting to review the unblinded data and consider early termination.
2. *Futility signal* – the conditional power falls below a pre-specified threshold (e.g., 10–20%). This triggers a DSMB review of whether to continue enrollment.
3. *Safety signal* – a secondary monitoring rule for adverse events crosses a safety boundary. This triggers immediate safety review regardless of the efficacy analysis schedule.

The DSMB retains full authority over all stopping decisions. The system merely ensures that evidence is not accumulating unexamined between scheduled meetings. This is particularly relevant for trials in which enrollment is fast relative to the DSMB meeting schedule: if 50 patients enroll between quarterly meetings, a group sequential design with $K = 5$ may miss the opportunity to stop the trial at the statistically optimal time.

A.5 The Diminishing Marginal Cost of More Analyses

A counterargument to high-frequency monitoring is that each interim analysis incurs administrative cost: data locks, database reconciliation, statistical programming, and DSMB coordination. This is true in the current paradigm but largely an artifact of it. If the monitoring system is automated – data are ingested continuously from electronic data capture, boundaries are pre-computed, and the test statistic is updated programmatically – then the marginal cost of the 21st analysis is approximately zero. The cost structure shifts from per-analysis overhead to a fixed upfront investment in infrastructure. For large trials where the cost per enrolled patient is high and the potential savings from early termination are substantial, this investment is easily justified.

A.6 Practical Recommendations

Based on the simulation results and the considerations above, we offer the following concrete recommendations:

1. **Default to $K = 20$ rather than $K = 5$.** For trials with electronically captured endpoints and turnaround times of less than one month, the spending function framework supports 20 planned analyses with no loss of type I error control and meaningful gains in expected sample size. The practical overhead is minimal if the monitoring system is automated.
2. **Use the O’Brien-Fleming spending function.** At high K , the O’Brien-Fleming spending function spends negligible α at early analyses, so the cost of ‘looking more often’ is almost entirely borne by a trivially small increase in the maximum sample size (typically less than 1% inflation relative to the fixed-sample design).
3. **Report bias-corrected estimates at every analysis.** The naive MLE should never be the sole estimate presented to a DSMB. At minimum, report the median unbiased

425 estimate and the repeated confidence interval alongside the naive values.

- 426 4. **Build the alerting system first.** The most impactful near-term investment is not in
427 novel statistical methodology but in software engineering: a system that ingests data,
428 updates the test statistic, evaluates it against pre-computed boundaries, and generates
429 structured alerts when thresholds are crossed. This system should produce a standard-
430 ized DSMB report at each triggered analysis, including the spending function status
431 (cumulative alpha spent), the bias-corrected estimate, conditional power, and a recom-
432 mendation for whether a DSMB meeting is warranted.
- 433 5. **Treat K as a design parameter to be optimized.** Rather than choosing K by
434 convention (the current default of 3–5), select K by explicitly trading off expected sam-
435 ple size savings against infrastructure cost. The diminishing-returns curve from the
436 simulation provides the empirical basis for this trade-off.

437 B References

438 B.1 Morris et al. (2019) ADEMP Compliance

439 We audited this simulation study against the reporting standards proposed by Morris, White,
440 and Crowther²⁸. The full audit is at `docs/morris-audit-2026-04-17.md`.

441 **Verdict:** Partially compliant.

442 Key gaps identified:

- 443 • No Monte Carlo SE on rejection rate, mean N, or bias in `sim_study.R`:94–108.
444 • `set.seed()` appears twice in the Rmd (main chunk and HF appendix), violating the
445 seed-once principle.
446 • `n_rep = 2000` not justified via a target MCSE.

447 A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

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